Tabs Block: How It Was Built and How to Author It

Audience: content authors, reviewers and developers working on this site in AEM Universal Editor.

About the screenshots: every image in this guide is the rendered result of sample authored content (the sandbox page `drafts/tabs.plain.html`), captured in the local preview. They show what visitors see for each set of authoring choices. They are not screenshots of the Universal Editor interface itself.

Status: the published-page behaviour shown here was tested in the local preview. Three editor-only behaviours still need confirming in your AEM author environment: switching tabs when you select content in the Content Tree, refreshing tabs after a panel edit, and showing *Auto Rotation Speed* only when rotation is on.

Contents

1. How the Tabs block was built
2. The key idea: a Tabs block plus Tab Panel sections
3. Authoring tabs step by step
4. Field reference
5. Variations, with settings and screenshots
6. Tablet and mobile behaviour
7. Accessibility: what is built in and what authors need to do
8. Tips, limits and troubleshooting
9. Testing locally

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1. How the Tabs block was built

1.1 Requirements

The goal was a reusable **container** Tabs block where each tab is a drop zone that can hold **any** block (hero, cards, columns, forms, video, custom blocks) without code

changes when new blocks are added. Authors must be able to add, delete, reorder and rename tabs, choose a default tab, and do all of this in Universal Editor. The block also needed 10 visual variations, three mobile behaviours, deep linking, auto rotation and WCAG 2.2 AA accessibility.

1.2 Research and the main design decision

Edge Delivery Services has one hard rule that shapes the design: **a block cannot contain other blocks.** A block's content is rows and cells, not child blocks. The only container that can hold any block is a **section**.

Adobe's own reference example for Universal Editor tabs (pull request #69 in `adobe-rnd/aem-boilerplate-xwalk`) solves this by making each tab panel a section. This block follows that pattern:

Piece	What it is	What it holds
Tabs block	A normal block placed in a section	The heading, description and all tab settings
Tab Panel	A special section type placed right after it	The tab's title, ID, icon and so on, plus any blocks

When the page loads, the Tabs block finds the Tab Panel sections that directly follow it, builds the tab bar from their titles, and moves the panels inside itself. Because a Tab Panel uses the **same allowed-components list as a normal section**, any block added to the project later is automatically available inside tabs.

1.3 Build steps

1. **Content model.** I created two Universal Editor models:
   - `blocks/tabs/_tabs.json` : the Tabs block (heading, description, rotation speed, plus 9 option fields that become CSS classes).
   - `models/_tab-panel.json` : the Tab Panel section (title, ID, icon, badge, description, thumbnail, default active, disabled, analytics ID, style).

   I registered both in the editor's component lists, allowed Tab Panels on the page and Tabs inside sections, and regenerated the editor configuration with `npm run build:json` .

2. **Behaviour (`tabs.js`).** Finds the panels; builds the tab buttons, accordion triggers and dropdown; handles clicks, the keyboard, URL hashes, auto rotation and the `tabs:change` analytics event. In Universal Editor it also opens the tab

containing whatever you select in the Content Tree, and rebuilds itself when you edit a panel.

3. **Styling (`tabs.css`).** Design tokens (CSS variables) for colours, spacing and sizes; one rule group per tab style; themes that only swap tokens; mobile-first breakpoints at 600px and 900px.
4. **Sample content.** A sandbox page with 9 tab groups covering every variation, used for testing and for the screenshots in this guide.
5. **Testing.** I checked every variation at desktop (1280–1440px), tablet (768px) and mobile (375–390px), and tested the keyboard, deep links, rotation, the event payload, screen-reader names and horizontal overflow. Linting (`npm run lint`) passes.

1.4 Problems found in testing and fixed

Problem	Fix
Auto rotation scrolled the whole page back to the tab bar every few seconds	Only the tab bar scrolls sideways now; the page never moves
The "Icon" style class clashed with the site's global `.icon` rule, shrinking the block to 24px	Renamed the style's class to `icon-tabs`
Screen readers announced panels as "Design Shape the target experience. Popular"	Panels are now named by the tab title only
On the dark theme, card text was white on white cards	The Cards block now sets its own text colour
Vertical underline tabs were spaced too far apart	Tightened the spacing in vertical layout

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2. The key idea: a Tabs block plus Tab Panel sections

```
Page
├── Section ............................... normal section
│   ├── (optional) text above the tabs
│   └── Tabs block ........................ settings: style, alignment, mobile beh
├── Tab Panel  "Overview" .................. tab 1 (drop zone)
│   ├── Text, Image, Button
│   └── Columns block
├── Tab Panel  "Features" .................. tab 2 (drop zone)
│   └── Cards block
├── Tab Panel  "Support" ................... tab 3 (drop zone)
│   └── Hero block
└── Section ............................... the first non-panel section ends the t
```

Rules to remember:

- **Order = tab order.** The first Tab Panel after the Tabs section is tab 1, and so on.
- **Keep panels together.** A normal section between two panels ends the tab group, and any panels after it display as ordinary sections.
- **One tab group per Tabs block.** You can have several tab groups on a page; each Tabs block takes the panels that follow it.

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3. Authoring tabs step by step

3.1 Create a tab group

1. Open the page in **Universal Editor**.
2. Select the section where the tabs should appear (or add a new **Section**).
3. Click **Add** (+) and choose **Tabs** from the Blocks list.
4. With the Tabs block selected, fill in the properties panel on the right:
   - **Heading** and **Heading Type** (usually H2).
   - **Description** (optional).
   - **Tab Style**, **Orientation**, **Alignment**, **Mobile Behaviour**, **Theme Variant**.
   - Toggles: **Enable Icons**, **Deep Linking** (on by default), **Sticky Tabs**, **Auto Rotation**.
   - **Auto Rotation Speed** appears when Auto Rotation is on (3–30 seconds).
5. Select the **page** (or main area) in the Content Tree, click **Add** (+) and choose **Sections → Tab Panel**. Place it directly after the section that contains the Tabs block.

6. Repeat step 5 for each tab. Each new panel starts with the title "New Tab".

> Until at least one Tab Panel follows it, the Tabs block shows a dashed hint in the editor: *"No tabs yet. In the Content Tree, select Main, click Add (+) → Tab Panel, then drag it directly below this section."* Visitors never see this hint.

3.2 Set up each tab

Select a Tab Panel and set, in the properties panel:

- **Tab Title** (required): the text on the tab.
- **Tab ID**: the deep link name, e.g. `overview` gives `page#overview`. Leave it empty to use the title.
- Optional: **Icon**, **Badge**, **Description**, **Image Thumbnail**, **Default Active**, **Disabled**, **Analytics Identifier**.

3.3 Add content to a tab

Select the Tab Panel and click **Add** (+). Any block allowed in a normal section can be added: text, images, buttons, hero, cards, columns, fragments, and any block added in the future. You can add several blocks to one tab.

> Tip: when you select something inside a tab in the Content Tree, the canvas automatically switches to that tab so you can see what you're editing.

3.4 Manage tabs

I want to…	Do this
Add a tab	Add another **Tab Panel** next to the existing panels
Delete a tab	Select the Tab Panel and delete it
Reorder tabs	Drag Tab Panels up or down in the **Content Tree**
Rename a tab	Change the panel's **Tab Title**
Open a specific tab first	Turn on **Default Active** on that panel (the first active panel wins)
Hide a tab temporarily	Turn on **Disabled** (shown greyed out, can't be opened)
Link to a tab	Use `#tab-id` in a link, e.g. `/services#cloud`

4. Field reference

4.1 Tabs block

Field	Options	Effect
Heading / Heading Type	Text, H2–H4	Title above the tab bar. Also used as the tab bar's accessible name.
Description	Rich text	Supporting text under the heading
Tab Style	Standard, Underline, Pill, Card, Icon, Numbered, Image, Mega	Look of the tabs (see section 5)
Orientation	Horizontal, Vertical	Vertical puts tabs on the left on desktop
Alignment	Start, Center, End, Stretch	Position of the tab bar; Stretch makes tabs share the full width
Mobile Behaviour	Scrollable tabs, Accordion, Dropdown	How tabs behave below 600px
Theme Variant	Light, Dark, Brand	Colour scheme of the whole tab group
Enable Icons	On/Off	Shows each panel's icon on its tab
Deep Linking	On/Off (default On)	Updates the URL when a tab is chosen; opens tabs from `#tab-id` links
Sticky Tabs	On/Off	Keeps the tab bar visible while scrolling long panels
Auto Rotation	On/Off	Cycles through the tabs until the visitor interacts
Auto Rotation Speed	3–30 seconds (default 6)	Only shown when Auto Rotation is on

4.2 Tab Panel

Field	Required	Notes
Section Name	No	Label in the Content Tree only
Tab Title	**Yes**	Up to 60 characters
Tab ID	No	Lowercase letters, numbers and hyphens; used in `#links`
Icon	No	Icon file name from the site's `/icons` folder, e.g. `info`, `star`, `chat`
Badge	No	Up to 20 characters, e.g. "New"
Description	No	Up to 160 characters; shown on Card, Image and Mega styles
Image Thumbnail	No	Shown on the Image style
Default Active	No	Opens this tab on page load
Disabled	No	Tab is shown but cannot be opened
Analytics Identifier	No	Sent with the tab change event
Style	No	Same section styles as a normal section (e.g. Highlight)

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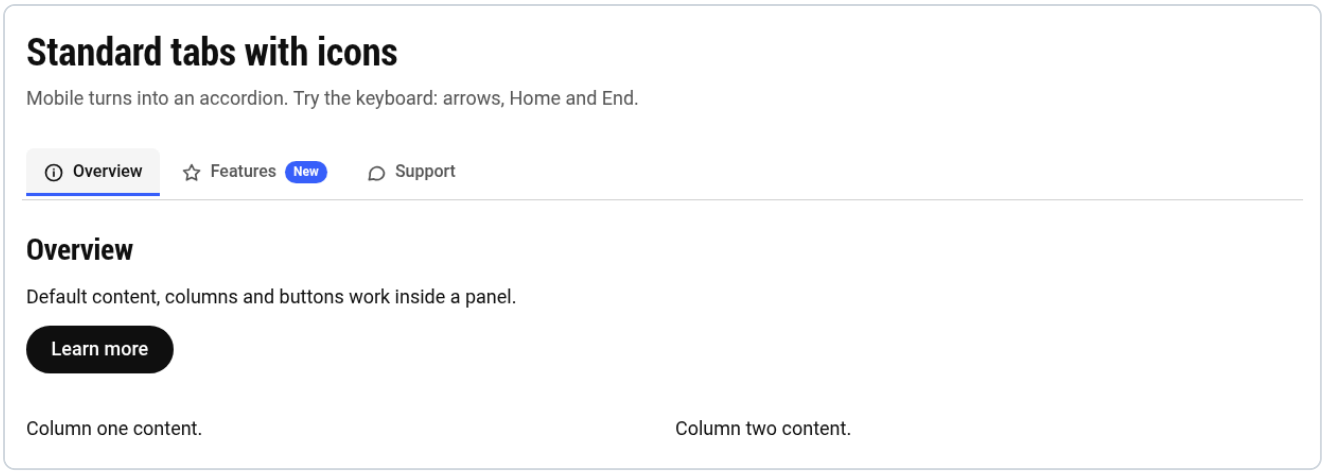
5. Variations, with settings and screenshots

Each variation below lists the settings to choose on the **Tabs block** and the fields that matter on the **Tab Panels**. Settings not listed can stay at their defaults.

5.1 Standard horizontal tabs

The default tab bar: the selected tab has a light background and an accent line. This example also has **Enable Icons** on and a badge on one tab.

Tabs block	Tab Panels
Tab Style **Standard** · Orientation **Horizontal** · Alignment **Start** · Mobile **Accordion** · Enable Icons **On**	Overview (`info` icon), Features (`star` icon, Badge "New"), Support (`chat` icon)



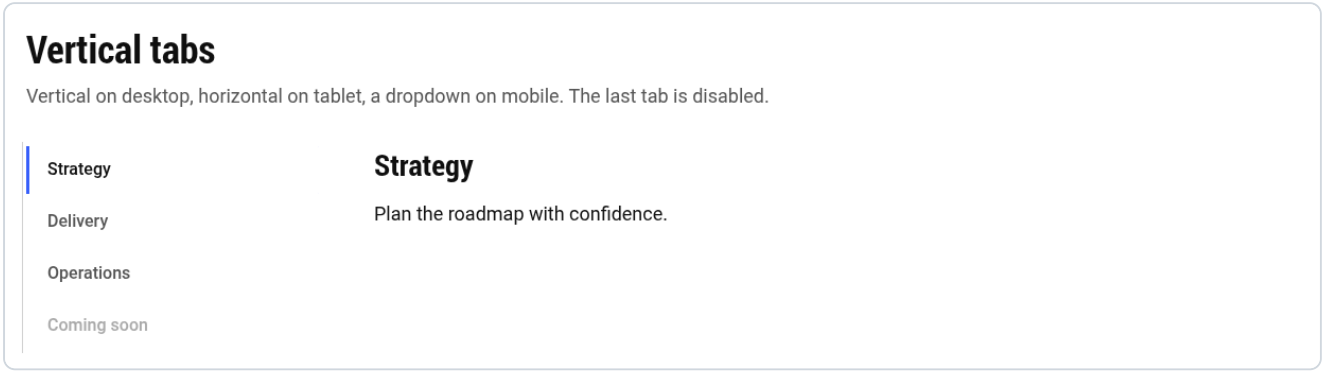
Keyboard focus is clearly visible when visitors use the arrow keys:



5.2 Vertical tabs

Tabs on the left and content on the right on desktop. Good for 4–8 tabs with longer titles. Turns horizontal on tablet and into a dropdown on mobile in this example. The last tab is **Disabled**.

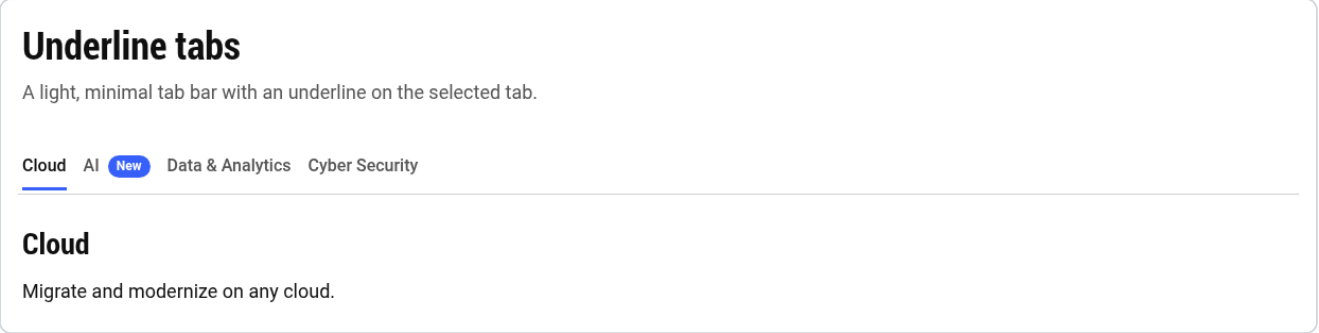
Tabs block	Tab Panels
Tab Style **Underline** · Orientation **Vertical** · Mobile **Dropdown**	Strategy, Delivery, Operations, Coming soon (**Disabled** on)



5.3 Underline tabs

A light, minimal tab bar with an underline under the selected tab. Good for in-page navigation between related topics.

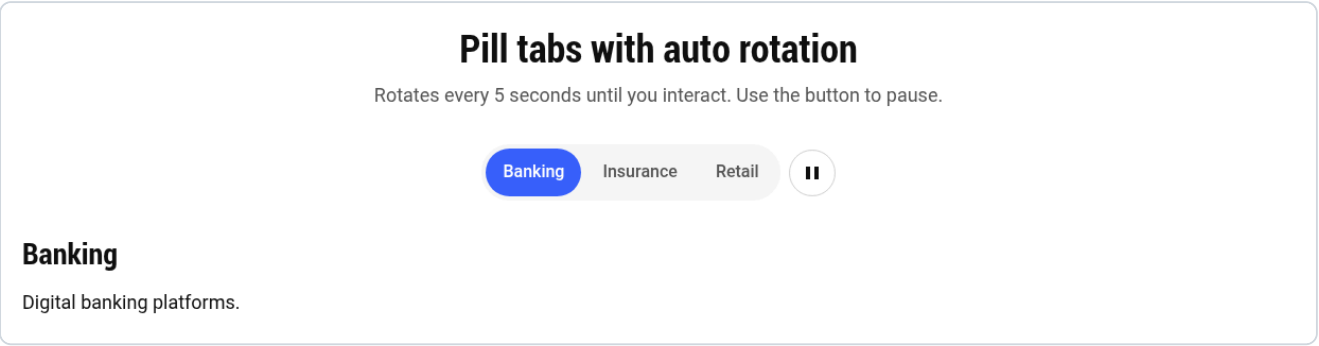
Tabs block	Tab Panels
Tab Style **Underline** · Orientation **Horizontal** · Mobile **Scrollable tabs**	Cloud, AI (Badge "New"), Data & Analytics, Cyber Security



5.4 Pill tabs (with auto rotation)

Rounded "segmented control" tabs; the selected tab is filled. This example is centred and rotates every 5 seconds, with a pause/play button.

Tabs block	Tab Panels
Tab Style **Pill** · Alignment **Center** · Auto Rotation **On** · Auto Rotation Speed **5**	Banking, Insurance, Retail

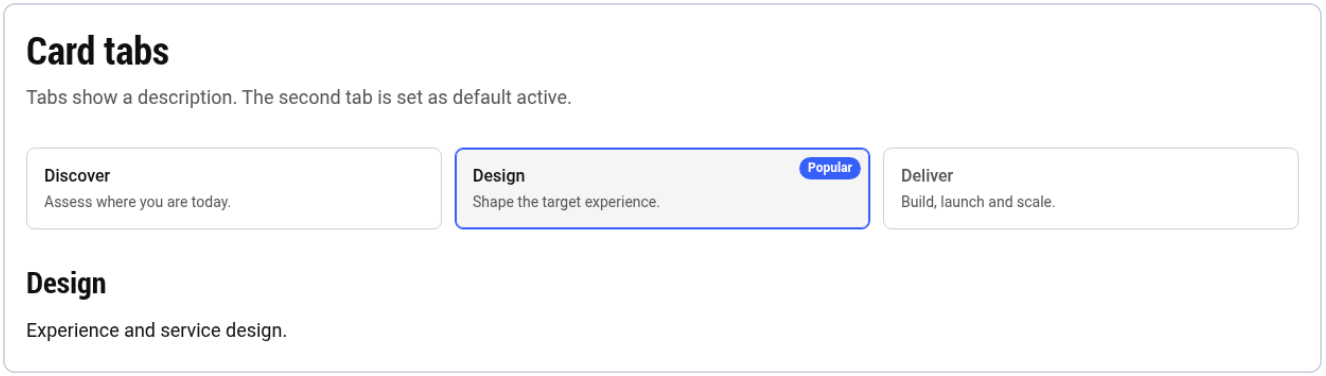


> Rotation stops for good as soon as the visitor clicks or uses the keyboard. It pauses while the pointer or keyboard focus is in the block, never runs for visitors who prefer reduced motion, and is switched off inside Universal Editor.

5.5 Card tabs

Each tab is a card with a title and short description. Good for 2–4 options that need explaining. In this example the second tab is the **Default Active** tab and has a badge.

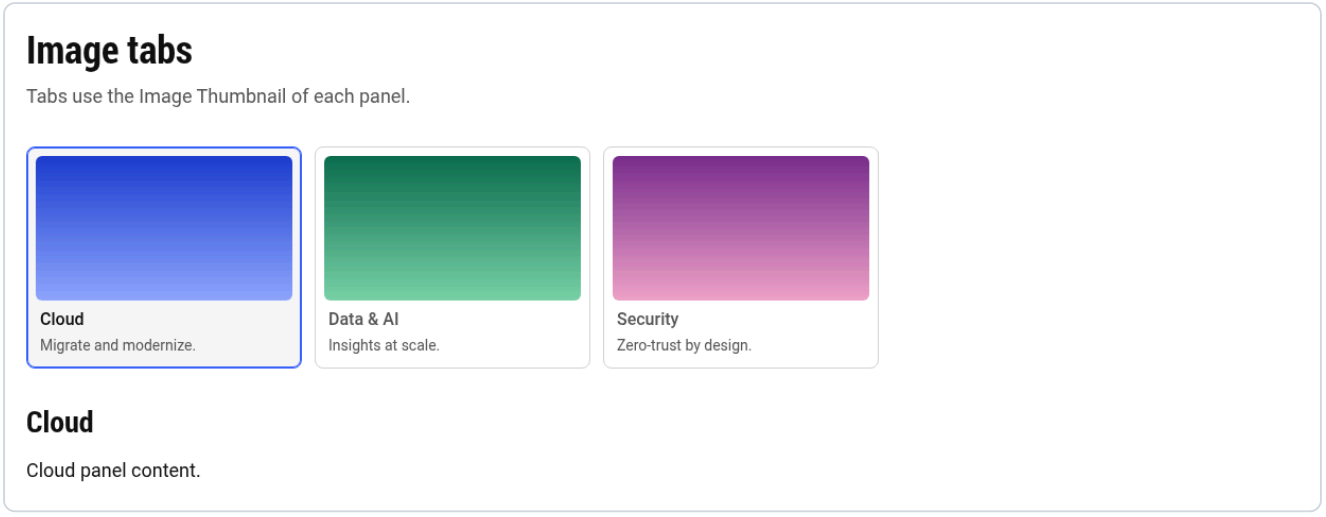
Tabs block	Tab Panels
Tab Style **Card** · Alignment **Stretch**	Each panel has a **Description**; Design has **Default Active** on and Badge "Popular"



5.6 Image tabs

Each tab shows a thumbnail image, title and description. Good for visual choices such as products or industries.

Tabs block	Tab Panels
Tab Style **Image**	Each panel has an **Image Thumbnail** and **Description**

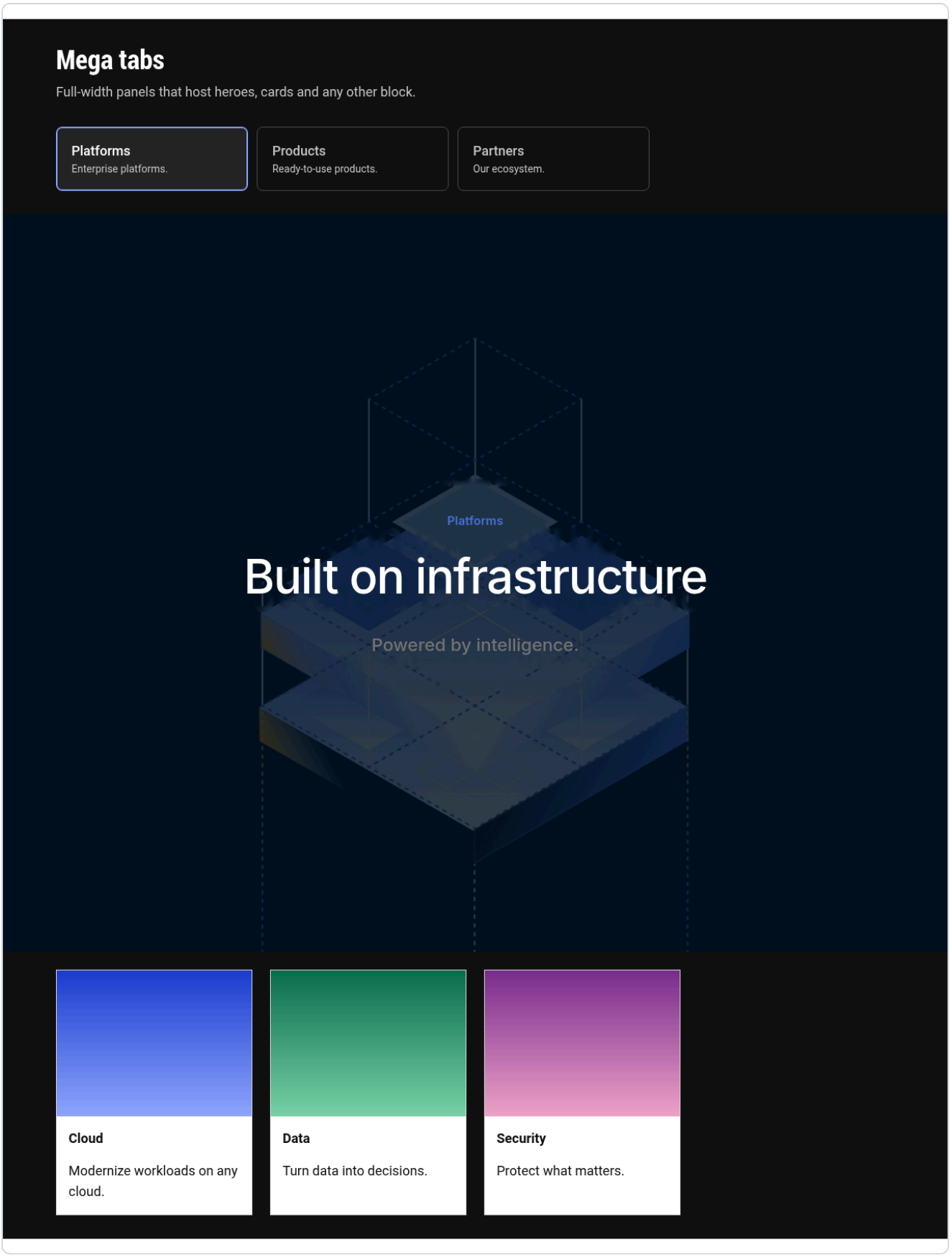


> Thumbnails are decorative (the tab title already describes the tab), so they don't need alt text.

5.7 Mega tabs

Large card-style tabs over **full-width panels**, so full-bleed blocks like a Hero span the page. Use this for major landing-page sections. This example uses the **Dark** theme; the first panel holds a Hero block followed by a Cards block.

Tabs block	Tab Panels
Tab Style **Mega** · Theme **Dark** · Mobile **Accordion**	Platforms (Hero + Cards), Products (Text + Columns), Partners (Text); each with a **Description**

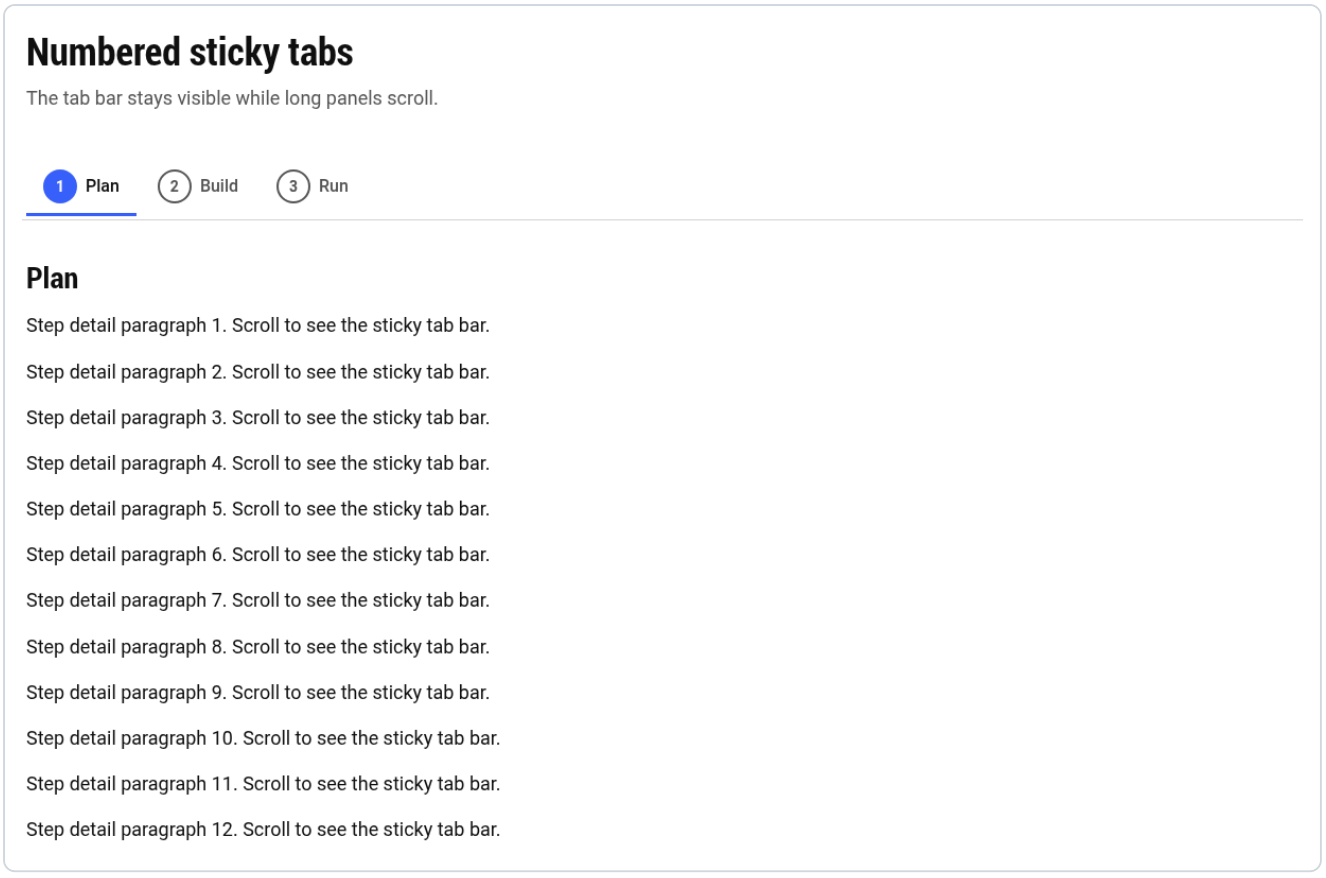


Mega tabs can hold any block: Hero, Cards, Columns, Fragment, forms, video, and custom blocks added later.

5.8 Numbered tabs (with sticky tabs)

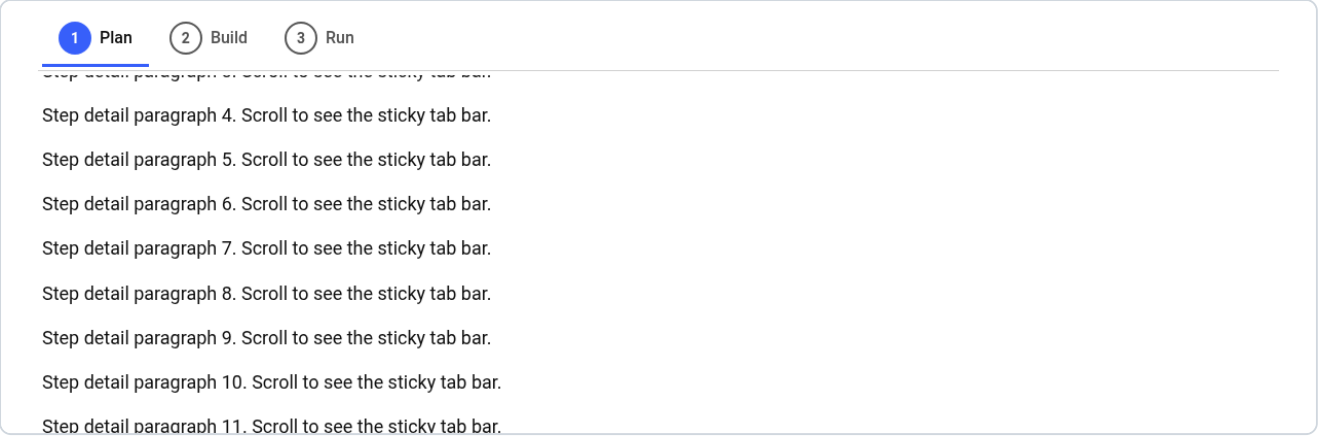
Tabs with step numbers. Good for processes such as Plan, Build, Run. This example also has **Sticky Tabs** on.

Tabs block	Tab Panels
Tab Style **Numbered** · Sticky Tabs **On**	Plan, Build, Run



5.9 Sticky tabs

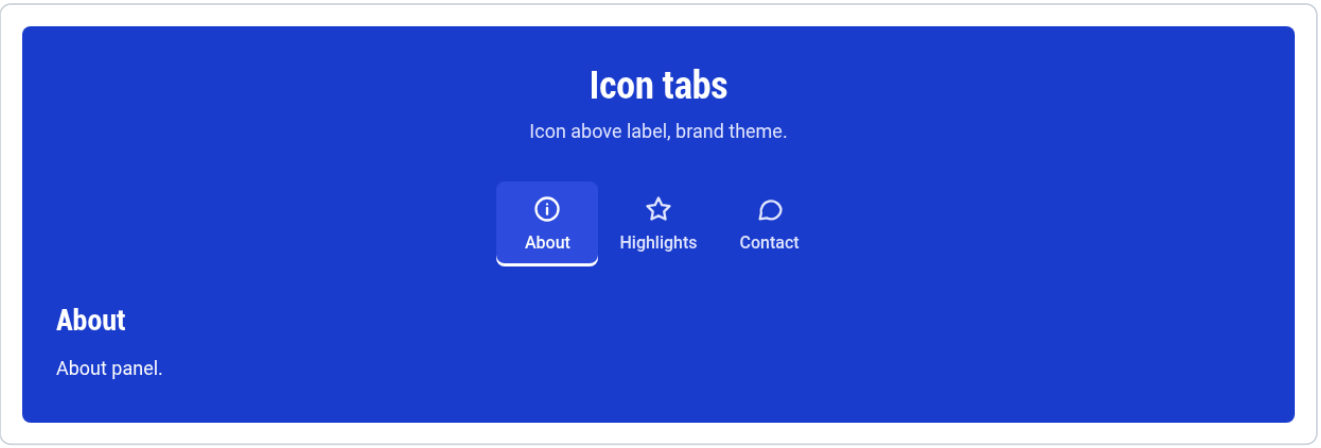
Any style can be sticky. While the visitor scrolls through a long panel, the tab bar stays pinned to the top (below the site header):



5.10 Icon tabs

The icon sits above the label. This example uses the **Brand** theme and is centred.

Tabs block	Tab Panels
Tab Style **Icon** · Alignment **Center** · Theme **Brand**	About (`info`), Highlights (`star`), Contact (`chat`)



> **Icon** (style) stacks the icon above the label. **Enable Icons** (toggle) adds icons next to the label on any other style, as in 5.1.

5.11 Quick picker

If you need…	Choose
A simple tab bar	Standard or Underline
A compact switch between 2–4 short options	Pill
Tabs that explain each option	Card
Visual choices	Image
Full-width page sections with heroes	Mega
Step-by-step content	Numbered
Many tabs or long titles	Orientation **Vertical**
Long panels	**Sticky Tabs** on
A showcase that cycles on its own	**Auto Rotation** on

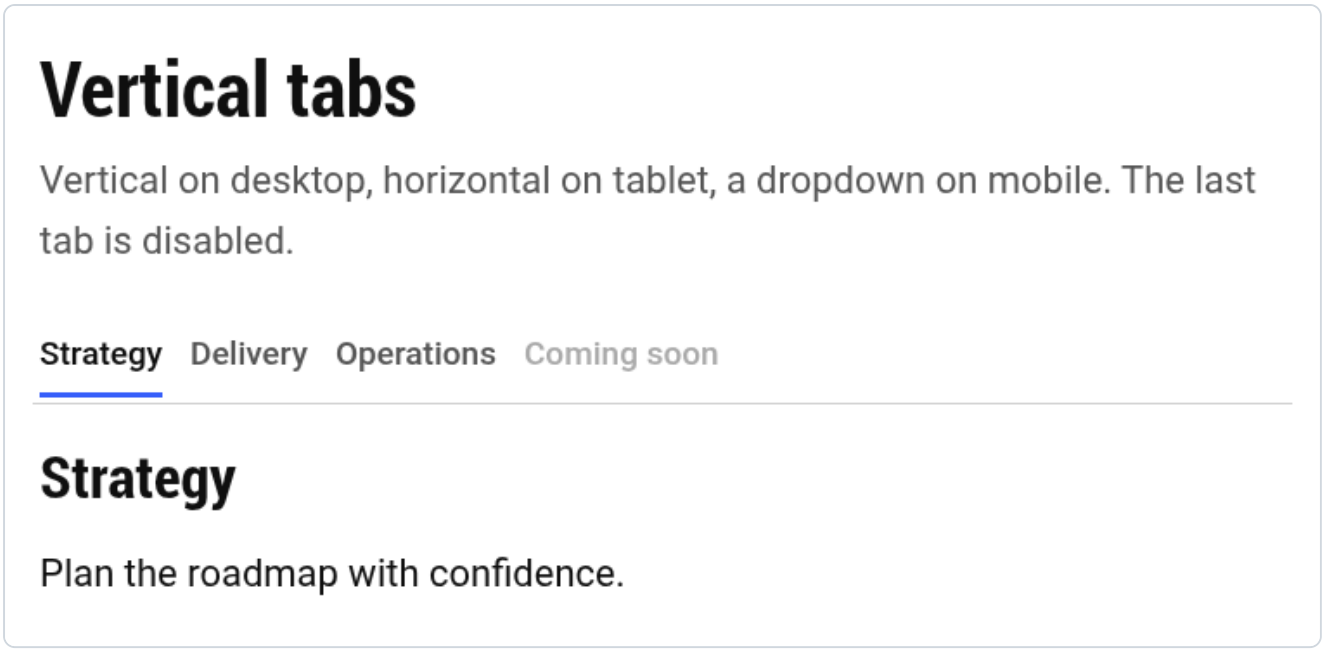
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6. Tablet and mobile behaviour

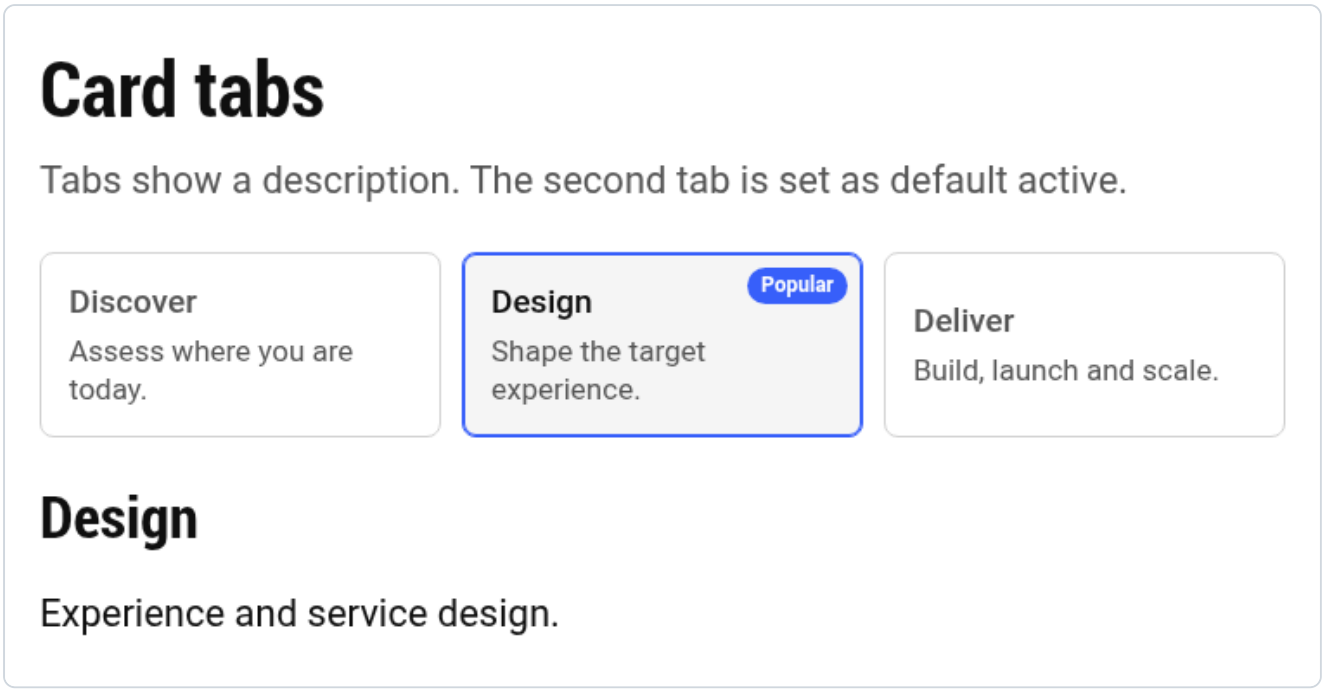
6.1 Tablet (600–899px)

Tabs are always horizontal on tablet. If they don't fit, the tab bar scrolls sideways (swipe), and the selected tab is kept in view.

Vertical tabs become horizontal on tablet:



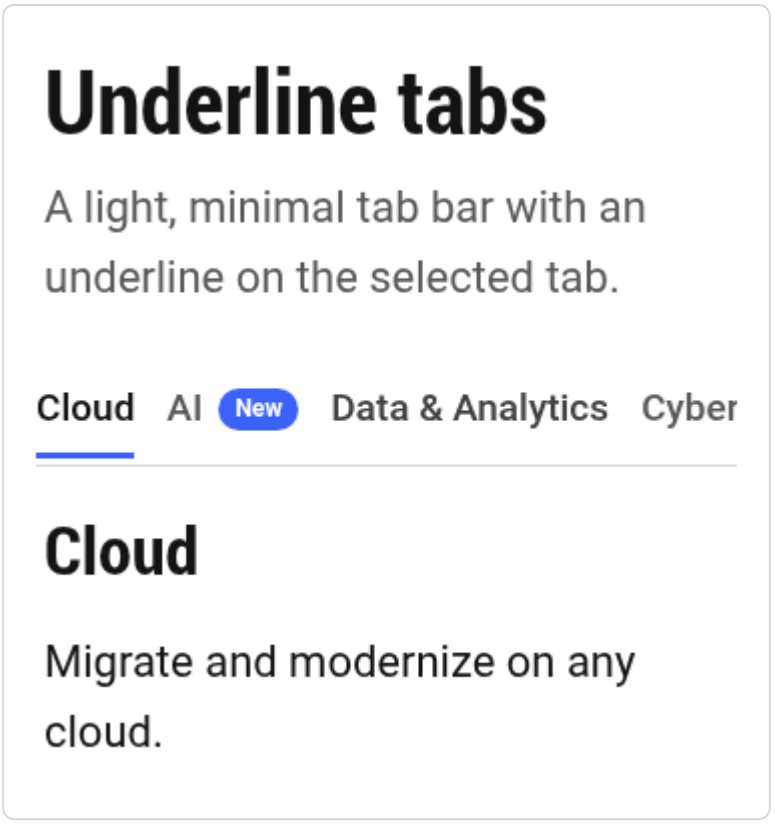
Card tabs scroll sideways when they don't fit:

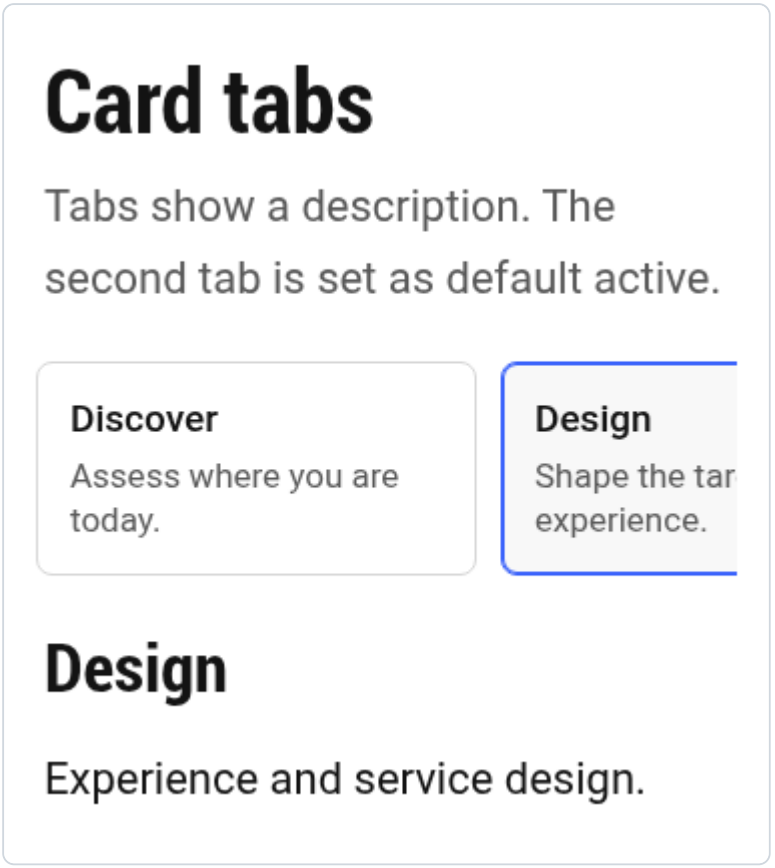


6.2 Mobile (below 600px): author's choice

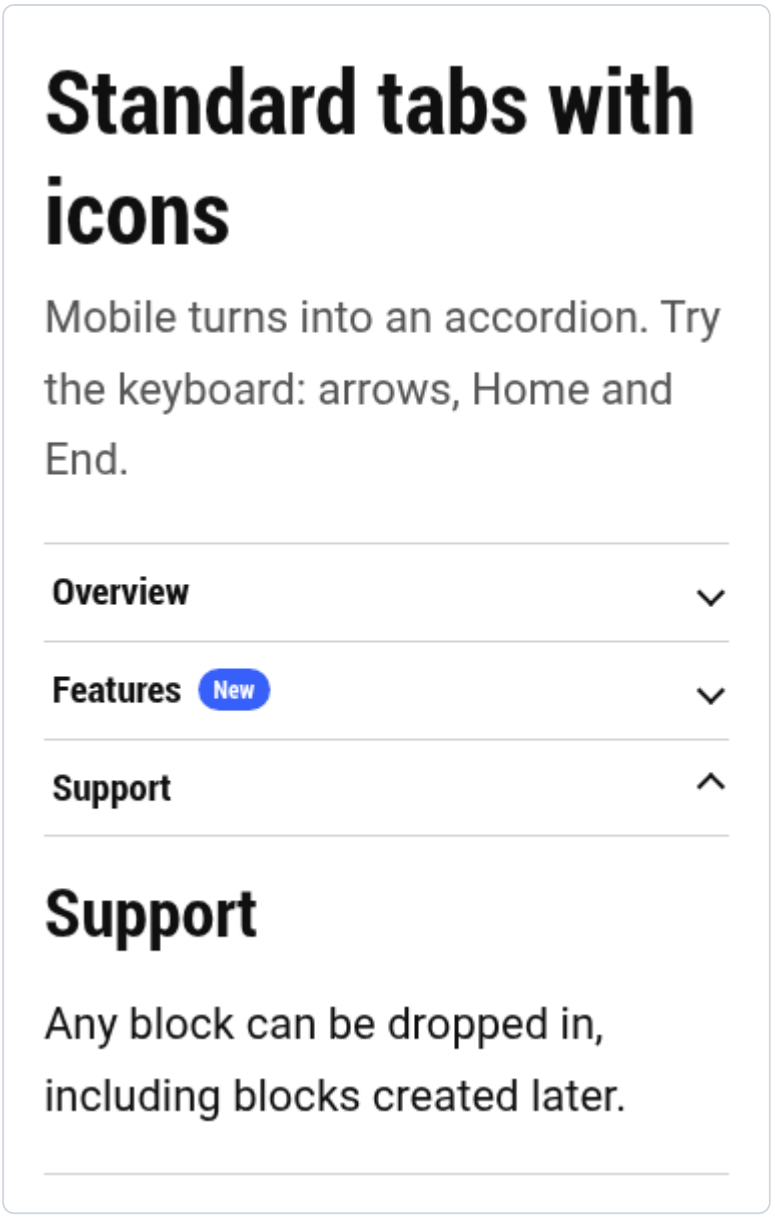
Choose **Mobile Behaviour** on the Tabs block:

Scrollable tabs: the tab bar stays and scrolls sideways.

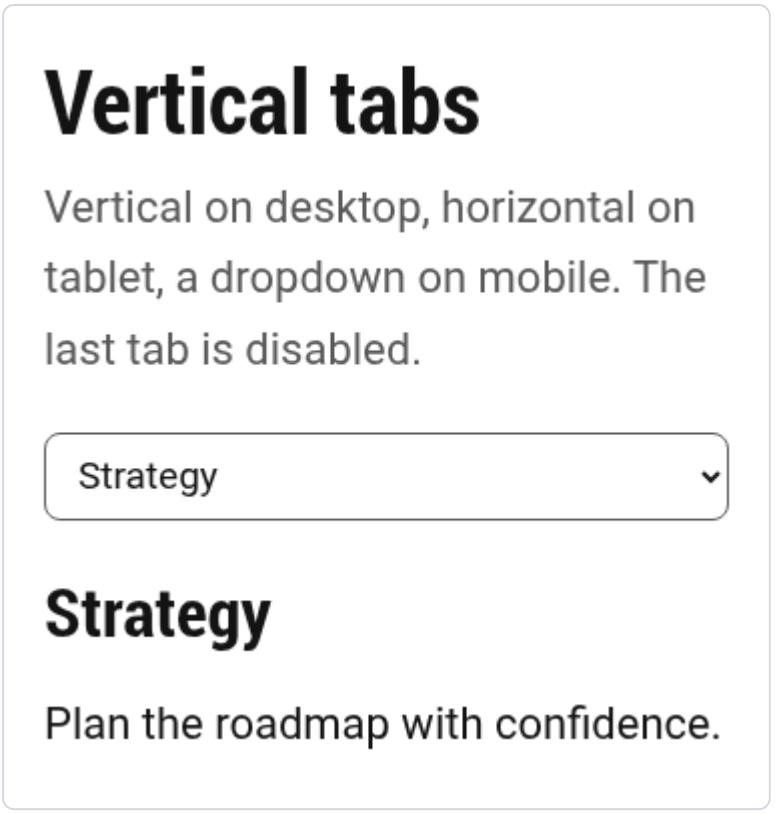




Accordion: each tab becomes a collapsible heading. One section is open at a time, and visitors can close them all.



Dropdown: a select menu replaces the tab bar. Disabled tabs can't be chosen.



Mobile Behaviour	Best for
Scrollable tabs	2–5 short titles
Accordion	Long content, many tabs, or Mega tabs
Dropdown	Many tabs with short panels

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7. Accessibility: what is built in and what authors need to do

Built in:

- Correct tab semantics for screen readers: tab list, tabs (with selected state) and tab panels, each linked to the other.
- Keyboard: **Tab** moves into the tab bar, ← → ↑ ↓ move between tabs, **Home**/**End** jump to the first or last tab. Disabled tabs are skipped.
- Visible focus outline on every interactive element.
- Tabs are at least 44px tall for easy tapping.
- The accordion and dropdown modes use their own correct semantics on mobile.
- Auto rotation has a pause/play button, pauses on hover and focus, and respects reduced-motion settings.

Authors should:

- Always give the Tabs block a **Heading**. It names the tab bar for screen-reader users.
- Write short, unique **Tab Titles** that describe each panel's content.
- Keep headings inside panels in order: if the Tabs heading is H2, use H3 inside panels.
- Add **alt text** to meaningful images inside panels (tab thumbnails don't need it).
- Use **Auto Rotation** sparingly, and only for non-essential showcase content.

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8. Tips, limits and troubleshooting

Situation	What's happening / what to do
I added a Tabs block but no tabs appear	Add **Tab Panel** sections directly after the section containing the Tabs block
A tab is missing from the tab bar	Make sure no normal section sits between the panels, and that the panel has a **Tab Title**
Adding the Tabs block didn't create Tab 1–3	Universal Editor inserts one component at a time, so add the panels yourself (each starts as "New Tab")
I want tabs inside a tab	Not supported: a Tabs block inside a Tab Panel shows a notice in the editor and is ignored on the live page
The wrong tab opens on page load	Check **Default Active** (the first panel with it on wins). A `#tab-id` in the URL takes priority
A deep link doesn't open the tab	Check the panel's **Tab ID** matches the link, e.g. `#cloud`
An icon doesn't show	**Enable Icons** must be on (or use the Icon style), and the icon name must match a file in `/icons`
Auto rotation doesn't run in the editor	Expected: it is switched off in Universal Editor and for visitors who prefer reduced motion

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9. Testing locally

The sandbox page used for the screenshots is in the repository at `drafts/tabs.plain.html`. It is excluded from the published site. To view it:

```
npx aem up --html-folder drafts --port 3001
# then open http://localhost:3001/drafts/tabs
```

Developer details (decorated HTML, JavaScript and CSS architecture, extensibility) are in `blocks/tabs/README.md`.